

MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 10

1. *Which of the following matrices are diagonalizable?*

(1) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Yes, in this case $\text{Nul}(A) = E_0(A)$(the eigenspace of A corresponding to $\lambda = 0$) $= \mathbb{R}^3$. Thus any basis of $\mathbb{R}^3$ will be an eigenbasis for A with eigenvalue 0.

(2) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$

Yes, since A has 3 distinct real eigenvalues $\lambda = 1, 4, 6$ then A will have an eigenbasis. The eigenbasis will be $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ where $\mathbf{v}_1$ is an eigenvector corresponding to the eigenvalue $\lambda = 1$, $\mathbf{v}_2$ is an eigenvector corresponding to the eigenvalue $\lambda = 4$, and $\mathbf{v}_3$ is an eigenvector corresponding to the eigenvalue $\lambda = 6$.

(3) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

No, A will not have an eigenbasis. There is only one eigenvalue $\lambda = 1$ with (algebraic) multiplicity 2. However, a calculation shows that $\text{Nul}(A - 1I)$ has dimension 1, so we can only find one linearly independent eigenvector for A. In order for A to have an eigenbasis, we would need to be able to find two linearly independent eigenvectors for A (since A is 2×2).

(4) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 2 & 3 \end{bmatrix}$

No. First we compute the eigenvalues via the characteristic polynomial:

$$p(\lambda) = \begin{vmatrix} 2-\lambda & 1 & 0 & 0 \\ 0 & 2-\lambda & 0 & 0 \\ 0 & 0 & 3-\lambda & 2 \\ 0 & 0 & 2 & 3-\lambda \end{vmatrix} = (2-\lambda)^2 \begin{vmatrix} 3-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} =$$

$$= (2-\lambda)^2((3-\lambda)^2-4) = (\lambda-2)^2(\lambda^2-6\lambda+5) = (\lambda-2)^2(\lambda-5)(\lambda-1)$$

The eigenvalues are $\lambda = 1, 2, 2, 5$. Since the (algebraic) multiplicity of $\lambda = 1$ and $\lambda = 5$ are both 1, we know that $\dim \mathrm{Nul}(A-5I) = \dim \mathrm{Nul}(A-1I) = 1$. Thus the only possible obstruction to the existence of an eigenbasis is $\dim \mathrm{Nul}(A-2I)$. If $\dim \mathrm{Nul}(A-2I) = 2$, then A has an eigenbasis, whereas if $\dim \mathrm{Nul}(A-2I) = 1$ then A does not have an eigenbasis. Let's compute $\dim \mathrm{Nul}(A-2I)$:

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3, R_3 \leftrightarrow R_4} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 3 pivots so $\mathrm{rank}(A-2I) = 3$ and so $\dim \mathrm{Nul}(A-2I) = 1$. Thus A does not have an eigenbasis. To summarize, A does not have an eigenbasis because we are only able to find 3 linearly independent eigenvectors. $\lambda = 1$ and $\lambda = 5$ both each contribute one linearly independent eigenvector, and $\lambda = 2$ also contributes only one linearly independent eigenvector. $\lambda = 2$ would have to give us two linearly independent eigenvectors in order for A to be diagonalizable (to get a grand total of 4 linearly independent eigenvectors) but it does not.

(5) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$

Yes, it is easy to see (since A is a diagonal matrix) that the usual standard basis of $\mathbb{R}^4$ will be an eigenbasis for A with respect to the eigenvalues $\lambda = 1, 2, 3, 3$.

2. *True/False. Give reasons why! Below A and B are matrices of an appropriate size.*

(1) If A is row equivalent to B, then they have the same eigenvalues.

False. For instance

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

are row equivalent, and 2 is an eigenvalue of B and is not an eigenvalue of A.

(2) If A is diagonalizable, then A^2 is diagonalizable.

True. If $A = PDP^{-1}$, then $A^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}$, therefore, P is also an eigenbasis of A^2.

(3) If all the eigenvalues of the matrix A are zero, then A is the zero matrix.

False. Consider $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

(4) If A is an $n \times n$ matrix with n different eigenvalues, then A is diagonalizable.

True. Since all the eigenvectors for different eigenvalues are linearly independent, n eigenvectors for different eigenvalues form an eigenbasis.

(5) Every triangular matrix has an eigenbasis.

False $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. The eigenspace with eigenvalue 0 is one dimensional.

3. *Give a formula for A^n for the following matrix A (Hint: use diagonalization):*

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$$

First we find the eigenvalues:

$$p(\lambda) = \begin{vmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{vmatrix} = \lambda^2 - 6\lambda + 5 = (\lambda - 5)(\lambda - 1)$$

Thus the eigenvalues are $\lambda = 1, 5$. The corresponding eigenvectors are $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Next we diagonalize:

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} -1/4 & 1/4 \\ 3/4 & 1/4 \end{bmatrix}$$

Taking powers yields:

$$A^n = \begin{bmatrix} -1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5^n \end{bmatrix} \begin{bmatrix} -1/4 & 1/4 \\ 3/4 & 1/4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 3 \cdot 5^n + 1 & 5^n - 1 \\ 3(5^n - 1) & 5^n + 3 \end{bmatrix}$$

4. (a) *Let A and B be $n \times n$ matrices. If B is an invertible matrix, does $B^{-1}AB$ always have the same eigenvalues as A?*

Yes, because they have the same characteristic polynomial:

$$p_{B^{-1}AB}(\lambda) = \det(B^{-1}AB - \lambda I) = \det(B^{-1}AB - \lambda B^{-1}IB) = \det(B^{-1}AB - B^{-1}\lambda IB)$$
$$= \det(B^{-1}(A - \lambda I)B) = \det(B^{-1}) \det(A - \lambda I) \det(B) = \det(A - \lambda I) = p_A(\lambda)$$

In the above calculation we used several basic properties of matrix and determinant arithmetic, try to justify each step for yourself!

(b) (Challenge problem) If A is an $n \times n$ matrix, and the eigenspace for $\lambda_1 \in \mathbb{R}$ has dimension d, show that the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$ of A contains $(\lambda - \lambda_1)^d$ as a factor. (This is why when λ_1 with an eigenvalue of A with multiplicity m, its eigenspace cannot have dimension bigger than m.)

We will show this by "partially diagonalizing" A and then applying part (a).
Let $\mathbf{v}_1, \ldots, \mathbf{v}_d \in \mathbb{R}^n$ be d linearly independent eigenvectors of A corresponding to the eigenvalue λ_1 (so $\{\mathbf{v}_1, \ldots, \mathbf{v}_d\}$ is a basis for the eigenspace of A corresponding to λ_1). Now complete $\{\mathbf{v}_1, \ldots, \mathbf{v}_d\}$ to a full basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_d, \mathbf{v}_{d+1}, \ldots, \mathbf{v}_n\}$ of $\mathbb{R}^n$ (it doesn't matter how you find the $n-d$ new vectors $\mathbf{v}_{d+1}, \ldots, \mathbf{v}_n$ and in general they probably won't be eigenvectors for other eigenvalues, it doesn't matter here...).
Now consider the $n \times n$ matrix

$$P = [\mathbf{v}_1 \cdots \mathbf{v}_d \ \mathbf{v}_{d+1} \cdots \mathbf{v}_n]$$

whose columns are the vectors from our basis of $\mathbb{R}^n$ (and the first d columns are eigenvectors corresponding to λ_1). P is invertible since its columns are linearly independent. Also, let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis for $\mathbb{R}^n$.
Let's consider the product $P^{-1}AP$. We compute

$$P^{-1}AP = P^{-1}A[\mathbf{v}_1 \cdots \mathbf{v}_d \ \mathbf{v}_{d+1} \cdots \mathbf{v}_n] = P^{-1}[A\mathbf{v}_1 \cdots A\mathbf{v}_d \ A\mathbf{v}_{d+1} \cdots A\mathbf{v}_n]$$

where we use the column-method for the product AP. However, we can simplify this slightly in the first d columns using that for $1 \leq i \leq d$, $\mathbf{v}_i$ is an eigenvector for A for eigenvalue λ_1 (so $A\mathbf{v}_i = \lambda_1 \mathbf{v}_i$ for $1 \leq i \leq d$):

$$\begin{aligned} P^{-1}AP &= P^{-1}[\lambda_1\mathbf{v}_1 \cdots \lambda_1\mathbf{v}_d \ A\mathbf{v}_{d+1} \cdots A\mathbf{v}_n] \\ &= [P^{-1}\lambda_1\mathbf{v}_1 \cdots P^{-1}\lambda_1\mathbf{v}_d \ P^{-1}A\mathbf{v}_{d+1} \cdots P^{-1}A\mathbf{v}_n] \\ &= [\lambda_1 P^{-1}\mathbf{v}_1 \cdots \lambda_1 P^{-1}\mathbf{v}_d \ P^{-1}A\mathbf{v}_{d+1} \cdots P^{-1}A\mathbf{v}_n] \\ &= [\lambda_1\mathbf{e}_1 \cdots \lambda_1\mathbf{e}_d \ P^{-1}A\mathbf{v}_{d+1} \cdots P^{-1}A\mathbf{v}_n] \end{aligned}$$

where in the last step we use that $P^{-1}\mathbf{v}_i = \mathbf{e}_i$ since $\mathbf{v}_i$ is the ith column of P. Note that this shows that the matrix $P^{-1}AP$ looks like a diagonal matrix in the first d columns with λ_1 on the diagonal (and in the last $n-d$ columns it doesn't matter what $P^{-1}AP$ looks like).
Computing the characteristic polynomial of $P^{-1}AP$ (say, by cofactor expansion recursively using always the first column) shows that

$$p_{P^{-1}AP}(\lambda) = (\lambda_1 - \lambda)^d q(\lambda)$$

where $q(\lambda)$ is some polynomial in λ of degree $n-d$. Thus $(\lambda - \lambda_1)^d$ is a factor of $p_{P^{-1}AP}(\lambda)$. In part (a) we showed that $p_{P^{-1}AP}(\lambda) = p_A(\lambda)$ and so $(\lambda - \lambda_1)^d$ is also a factor of $p_A(\lambda)$.

5. *Consider the vectors* $\mathbf{u}$ *and* $\mathbf{v}$ *in* $\mathbb{R}^3$*:*

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Decompose $\mathbf{v}$ *into a sum of two vectors, one which points in the same direction as* $\mathbf{u}$ *and one that is orthogonal to* $\mathbf{u}$*.*

We need to find c in $\mathbb{R}$ such that $\mathbf{v} - c\mathbf{u}$ is orthogonal to $\mathbf{u}$ (because $c\mathbf{u}$ is in the direction of $\mathbf{u}$ and $\mathbf{v} = c\mathbf{u} + (\mathbf{v} - c\mathbf{u})$). Thus we have find c such that

$$(\mathbf{v} - c\mathbf{u}) \cdot \mathbf{u} = 0.$$

Solving for c we get

$$c = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} = \frac{-4}{7}.$$

Thus

$$\mathbf{v} = c\mathbf{u} + (\mathbf{v} - c\mathbf{u}) = \underbrace{\frac{-4}{7}\begin{bmatrix}1\\2\\-3\end{bmatrix}}_{\text{in the direction of } \mathbf{u}} + \underbrace{\left(\begin{bmatrix}1\\3\\5\end{bmatrix} - \frac{-4}{7}\begin{bmatrix}1\\2\\-3\end{bmatrix}\right)}_{\text{orthogonal to } \mathbf{u}} = \begin{bmatrix}\frac{-4}{7}\\\frac{-8}{7}\\\frac{12}{7}\end{bmatrix} + \begin{bmatrix}11/7\\29/7\\23/7\end{bmatrix}.$$

6. *Let* $\mathbf{v}$ *be a nonzero vector in* $\mathbb{R}^n$*. A* ***reflector*** *is a matrix* R *such that* $R\mathbf{x} = -\mathbf{x}$ *if* x *is a scalar multiple of* $\mathbf{v}$*, and* $R\mathbf{x} = \mathbf{x}$ *if* $\mathbf{v} \cdot \mathbf{x} = 0$*.*

(1) Someone tells you that '*When applying a reflector, the orthogonal complement of* span($\mathbf{v}$) acts as a mirror.' Can you explain what that person meant? Draw a picture!

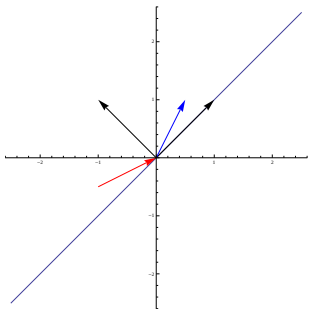
Let's consider an example. Let $n = 2$. Suppose that $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. The orthogonal complement of span($\mathbf{v}$) is span($\begin{bmatrix} 1 \\ 1 \end{bmatrix}$). Take the vector $\mathbf{w} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$. We now calculate $R\mathbf{w}$. Observe that

$$\begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} -0.25 \\ 0.25 \end{bmatrix}}_{\text{in span}(\mathbf{v})} + \underbrace{\begin{bmatrix} 0.75 \\ 0.75 \end{bmatrix}}_{\text{in span}(\mathbf{v})^\perp}$$

Thus

$$R\begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = R\underbrace{\begin{bmatrix} -0.25 \\ 0.25 \end{bmatrix}}_{\text{in span}(\mathbf{v})} + R\underbrace{\begin{bmatrix} 0.75 \\ 0.75 \end{bmatrix}}_{\text{in span}(\mathbf{v})^\perp} = -\begin{bmatrix} -0.25 \\ 0.25 \end{bmatrix} + \begin{bmatrix} 0.75 \\ 0.75 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}.$$

To interpret this result, look at the following picture.



The blue vector is $\begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$, the red vector is $\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ (shifted such that it points to the origin). You can think of the red vector as a light beam going in the direction of $\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ that hits the mirror (ie. the orthogonal complement of span($\mathbf{v}$)) at the origin. The blue vector is than the reflection of this light beam. Observe that the law of reflection holds: the direction of incoming light (the incident ray/red vectors), and the direction of outgoing light (the reflected ray/blue vector) make the same angle with respect to the surface normal (ie. $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$).

(2) Show that $R = 2P - I$, where P is projection matrix for projection onto the orthogonal complement of span($\mathbf{v}$). (If you haven't learned about projections onto arbitrary subspaces, then just consider the case $n = 2$).

Let $\mathbf{w}_1, \ldots, \mathbf{w}_{n-1}$ be a basis of span($\mathbf{v}$)$^\perp$. Then $\mathbf{v}, \mathbf{w}_1, \ldots, \mathbf{w}_{n-1}$ is a basis of $\mathbb{R}^n$ (Why? Orthogonality implies linear independence for non-zero vectors!). In order to show that $R = 2P - I$ is the enough show $R\mathbf{v} = (2P - I)\mathbf{v}$ and $R\mathbf{w}_i = (2P - I)\mathbf{w}_i$ for each $i = 1, \ldots, n-1$. Why is this enough? Because every linear transformation (and hence every matrix) is uniquely determined by the image of a given basis. So let's check:

$$R\mathbf{v} = -\mathbf{v} = 0 - \mathbf{v} = 2P\mathbf{v} - \mathbf{v} = (2P - I)\mathbf{v},$$

because $P\mathbf{v} = 0$.

$$R\mathbf{w}_i = \mathbf{w}_i = 2\mathbf{w}_i - \mathbf{w_i} = 2P\mathbf{w}_i - \mathbf{w}_i = (2P - I)\mathbf{w}_i,$$

because $\mathbf{w}_i$ is in the orthogonal complement of span($\mathbf{v}$).

(3) Deduce from *(2)* that $R = I - \frac{2}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\mathbf{v}^T$.

In class we learnt that the projection matrix S for projecting onto $\text{span}(\mathbf{v})$ is $\frac{1}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\mathbf{v}^T$. We also learnt that if S is the projection matrix for projecting onto $\text{span}(\mathbf{v})$, then $I - S$ is the projection matrix for projection onto $\text{span}(\mathbf{v})^\perp$. Thus

$$R = 2P - I = 2(I - S) - I = I - 2S = I - \frac{2}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\mathbf{v}^T.$$

The matrix on the right is called the **Householder matrix**.

(4) Deduce from *(3)* that $R^T = R$ and $R^T R = I$.

Using properties of the transpose, we get

$$\begin{aligned} R^T &= \left(I - \frac{2}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\mathbf{v}^T\right)^T \\ &= I^T - \left(\frac{2}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\mathbf{v}^T\right)^T \\ &= I - \frac{2}{\mathbf{v}\cdot\mathbf{v}}\left(\mathbf{v}\mathbf{v}^T\right)^T \\ &= I - \frac{2}{\mathbf{v}\cdot\mathbf{v}}\left(\mathbf{v}^T\right)^T\mathbf{v}^T \\ &= I - \frac{2}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}\mathbf{v}^T = R. \end{aligned}$$

Moreover, since $P^2 = P$, we get

$$R^T R = RR = (2P - I)(2P - I) = 4P^2 - 2P - 2P + I = 4P - 4P + I = I.$$